CERTPATH PUBLISHING · FORMULA CHEAT SHEET

POSS Exam — Quick Reference

EEI Plant Operator Selection System · Mechanical · Reading · Math · Graphs

⚙️ MECHANICAL REASONING

Simple Machines

- **Lever:** $F_e \times d_e = F_l \times d_l$
- **Gear ratio:** T_{driven} / T_{driver}.
- **Pulley MA:** # supporting ropes.
- **Hydraulic:** $F_1/A_1 = F_2/A_2$.
- Bigger piston → more force, less travel.

Fluid Flow & Pumps

```
Q = V × A (flow = velocity × area)
```

- Smaller pipe → higher velocity.
- **Centrifugal pump laws:**
  flow ∝ speed · head ∝ speed² · power ∝ speed³
- **NPSH:** suction pressure needed to prevent cavitation.

🔥 PLANT SYSTEMS

Heat & Steam

- **Sensible heat:** temperature changes.
- **Latent heat:** phase change, no temp change.
- **Superheated steam:** above saturation temp; used in turbines.
- Heat always flows hot → cold.

Electricity for Operators

```
V = IR · P = VI
```

- Transformer: $V_p/V_s = N_p/N_s$.
- Step-up V = step-down I (power constant).
- 3-phase: more power per conductor.

🧮 MATH & GRAPHS

Percent & Ratio

```
% = part / whole × 100

Δ% = (new − old) / old × 100
```

- Ratio 3:2 = 3 parts per 2.

Basic Algebra

- Isolate the variable; do the same to both sides.
- Distribute, combine like terms, solve.
- Check by plugging back in.

Graphs & Tables

- Read both axes labels before answering.
- Identify peaks, valleys, trends.
- Line = change over time; bar = categories.

📖 **Reading comprehension:** read question first, then scan passage. Answer based only on what's in the text — outside knowledge is a trap. Eliminate extreme-language answers ("always", "never"). When two answers seem close, choose the one most directly supported by a specific line.